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Cognitive biases in individuals with chronic pain: interpretation, recall and recognition of pain/illness-related information

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ABSTRACT

Chronic Pain (CP) significantly impairs quality of life, making it critical to understand the psychological factors that contribute to its onset and persistence. This study investigated interpretation biases (IB) and memory biases (MB) in 153 individuals (CP = 77; non-pain controls = 76). Participants completed tasks assessing IB using ambiguous scenarios, and MB via surprise short-term recall and delayed recognition (after one month) of pain/illness-related information. Results indicated no significant group differences in IB or short-term memory bias recall, and no significant associations between IB, recall, and recognition in either group. However, signal detection analyses revealed that individuals with CP showed significantly reduced long-term memory bias recognition sensitivity for pain-related stimuli. These findings suggest that CP may selectively disrupt later-stage memory processes for pain-related information, supporting a temporally differentiated account of cognitive bias in CP. This highlights the need to refine existing theoretical models and frameworks.

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Chronic pain; cognitive biases; interpretation; memory; recognition

1. Introduction

Pain, as defined by the International Association for the Study of Pain, encompasses both sensory and emotional components associated with actual or potential tissue damage (Raja et al., 2020). It is categorised temporally, with duration of <3 months considered acute and duration of >3 months considered chronic (Raffaelli et al., 2021). Pain affects an estimated 35–50% of individuals in the United Kingdom (UK), a statistic anticipated to rise substantially in the future (Fayaz et al., 2016). Chronic pain (CP) imposes multifaceted burdens. Physically, it restricts functionality (Dagnino & Campos, 2022), disrupts sleep (Husak & Bair, 2020) and leads to medication-related side effects including nausea (Martel et al., 2015; McCracken et al., 2006). Psychologically, CP is associated with increased risk of depression, anxiety, cognitive impairments such as impaired executive function, and profound emotional distress including frustration, helplessness and hopelessness (Caes et al., 2021; Lumley et al., 2022; Maher & Gaffiero, 2025; Rogers & Farris, 2022). Socially, CP strains relationships, reduces social engagement and impairs work participation (McMurtry et al., 2020; Yong et al., 2022). Recognising these impacts highlights the necessity

of understanding the factors contributing to the onset and persistence of CP. This is crucial for developing effective pain management interventions as well as policies alleviating its societal burden.

Western medicine prioritises physical contributors to CP, yet a biopsychosocial approach (Engel, 1977) increasingly emphasises holistic, person-centred CP care. This approach considers biological, psychological and social factors interacting to shape CP development and maintenance (Turvill et al., 2026). Over the past three decades, accumulating evidence has highlighted psychological factors, including information processing biases such as interpretation bias (IB) and memory bias (MB) as pivotal in contributing to the aetiology and maintenance of CP (Pincus & Morley, 2001; Schoth & Lioffi, 2016; Schoth et al., 2020).

IB entails appraising ambiguous scenarios negatively or threateningly. For instance, individuals with chronic back pain might interpret slight discomfort during light lifting as indicative of worsening pain or injury. This then fosters anxiety, activity avoidance and further physical deterioration, perpetuating the CP cycle. MB refers to a memory bias characterised by the preferential

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encoding, consolidation, or retrieval of pain- or illness-related information (Kuhajda et al., 2002). For example, those with chronic migraines might recall predominantly severe episodes while downplaying manageable ones, amplifying perceived severity and thereby increasing anxiety. Understanding development and maintenance of these biases is therefore crucial in managing CP, and informing psychological (e.g. IB modification) and physical (e.g. biofeedback) targeted interventions.

Evidence supporting IB presence among individuals with CP has grown considerably in recent years. A meta-analysis involving 445 individuals with CP, 407 healthy controls (HCs) and 170 healthcare professionals, confirmed that individuals with CP are prone to interpreting ambiguous information as being pain/illness related (Schoth & Liossi, 2016). All seven studies included in that review demonstrated that individuals with CP are notably more likely to interpret ambiguous information in a pain/illness-related manner compared with HCs. Subset meta-analysis of data from four of these studies revealed a moderate effect size, highlighting the robustness of IB in individuals with CP. Subsequent studies have replicated these effects across various CP conditions including chronic fatigue syndrome, endometriosis and rheumatoid arthritis (Boyse et al., 2024; Hughes et al., 2017; Khatibi Tabatabaei, 2014; Pickup et al., 2023; Schoth et al., 2019). Moreover, O'Hara et al. (2022) observed moderate-to-large effect sizes for IB in populations with chronic headache/migraine, demonstrative of its pervasive nature and impact on cognitive processes in CP (O'Hara et al., 2022).

By contrast, the implications of MB in CP remain contentious. Early reviews suggested that individuals with CP recall more pain/illness-associated information (Pincus & Morley, 2001), yet a systematic review and meta-analysis found inconclusive evidence across 12 studies comparing individuals with CP and HCs (Schoth et al., 2020). While individuals with CP showed biased recall favouring sensory pain/illness-related words, no marked differences in the recall of affective pain- and depression-related words were observed. Similarly, research in recognition bias has also proved equivocal. For example, individuals with depression and CP have been found to display increased recognition of pain/illness-related words (Pincus et al., 1995). Conversely, impaired recognition performance has been identified in: (i) Patients with CP compared with HCs (Grisart et al., 2007); and (ii) populations subjected to pain experimentally (Forkmann et al., 2016). To sum, the exact nature of recall and recognition biases with regards to pain experiences is unclear. Such variation in findings indicates that MB in the context of CP is complex. Its nature may not be consistent across all individuals or pain conditions. Thus, additional

methodologically rigorous investigation is warranted to further elucidate the role of MB in CP perception.

Together, The Threat Interpretation Model (TIM) (Todd et al., 2015) and The Integrated Functional Contextual Framework (IFCF) (Van Ryckeghem et al., 2019), offer a theoretical lens highlighting how cognitive biases (CBs) could perpetuate CP. According to the IFCF, CBs are inter-related, dynamic and interact in ways that can exacerbate CP conditions. According to the TIM, potential pain processing typically begins with an initial stimulus, such as ambiguous bodily sensations (e.g. discomfort, tightness) or an external pain-related cue (e.g. seeing something associated with past pain). If this stimulus is deemed pain-related and threatening, the attention of individuals with CP is captured, resulting in a vigilance-avoidance attentional bias (AB). However, if the stimulus is categorised as non-pain related or non-threatening, normal attentional processing occurs (see Figure 1 below). Whilst these models are useful both have caveats. For example, it should be noted that the TIM does not integrate MBs, and the IFCF offers no specific, testable predictions regarding how cognitive biases interact over time.

Studies exploring IB and MB simultaneously remain sparse. Schoth et al. (2018) demonstrated IB and MB co-occurrence in chronic headache, with individuals with chronic headache providing more sensory pain/illness-related responses in a sentence generation task (IB) and also recalling more sensory pain/illness-related words in a surprise-free recall task compared to HCs (Schoth et al., 2018). However, in a subsequent study featuring the same population, Schoth et al. (2019) found evidence of IB but no evidence of MB (recall). Xu et al. (2024) recently explored the interrelationships between CB including IB and MB, in healthy participants using a new pain paradigm. This allowed IB and MB to be examined simultaneously in response to personally relevant stimuli (i.e. cues signalling actual pain), followed by the cold-pressor task to assess pain outcomes. Xu et al. found that participants exhibited IB and MB for conditioned stimuli (e.g. those paired with pain). However, no relationship between IB and MB was observed, questioning the IFCF. Given the IFCF highlights the interplay of pain/illness-related CBs, further research is needed to clarify individual and combined influences of IB and MB on pain experiences.

Thus, in the present study we assessed IB using ambiguous scenarios, and the two forms of MB—recall and recognition—in individuals with CP and NPCs (non-pain controls). Considering the aforementioned literature, it was hypothesised that compared with NPCs, individuals with CP would: (i) Interpret more ambiguous scenarios in a pain/illness-related manner (hypothesis 1); and (ii) recall more pain/illness-related solutions in a surprise short-term MB task (hypothesis 2). It was further

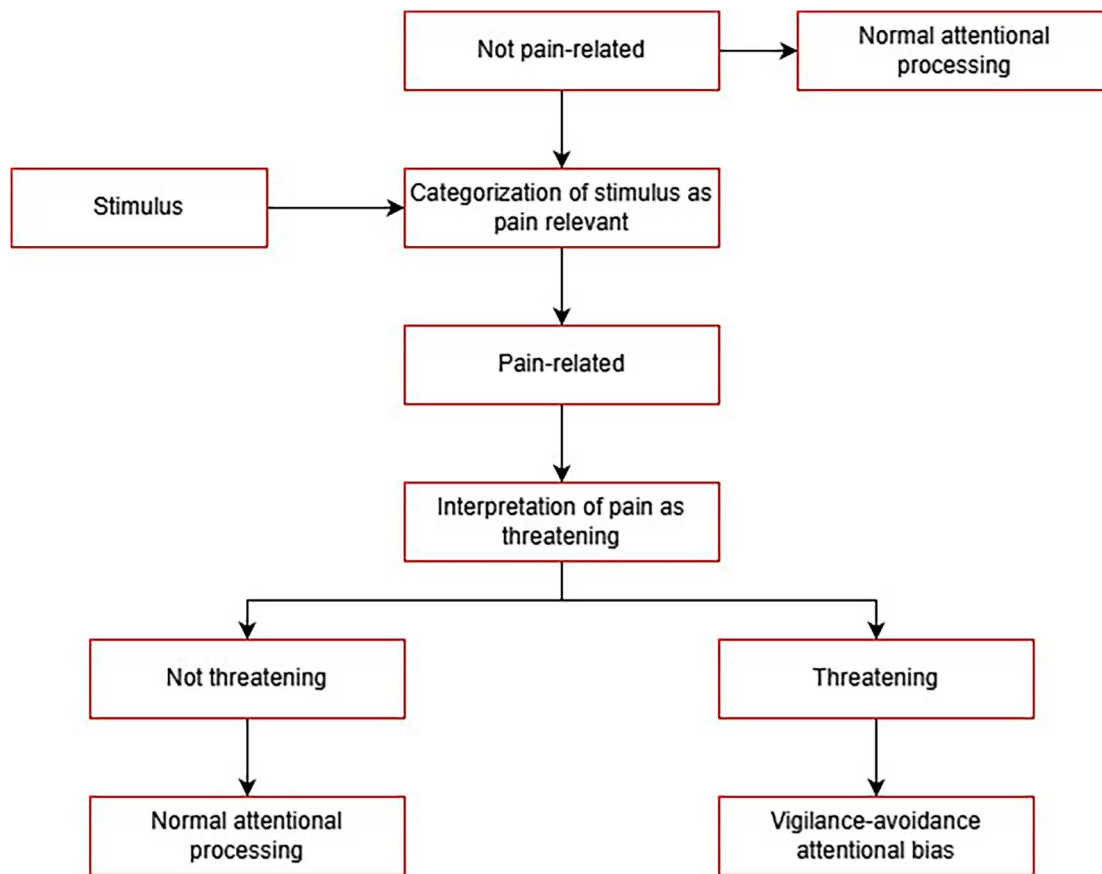


Figure 1. The threat interpretation model developed by Todd et al. (2015). Permission granted for use by author.

hypothesised that individuals with CP and NPCs would differ in recognition of pain/illness-related solutions presented 1 month later in a long-term MB task (hypothesis 3) and that there would be significant associations among the CB in individuals with CP (hypothesis 4).

2. Materials and methods

2.1. Design

The present study employed a mixed-measures design with three independent variables. The between-subjects variable was pain (individuals with CP vs. NPCs) determined by asking participants to confirm whether they have experienced pain for ≥ 12 weeks. The within-subjects variables included: (i) Scenario category (ambiguous pain vs. ambiguous neutral) specific to the IB task; and (ii) word type (pain/illness related vs. non-pain/non-illness related) specific to the MB tasks. The dependent variables were: (i) the proportion of pain/illness-related vs. non-pain/non-illness-related interpretations (interpretation bias task); the proportion of pain/illness-related vs. non-pain/non-illness-related solutions correctly recalled (surprise free recall task); and recognition accuracy indices expressed as proportions of hits (H), misses (M), false

alarms (FA) and correct rejections (CR) for pain/illness-related and non-pain/non-illness-related solutions (delayed recognition task). The current study was administered online to widen participant access considering the vulnerable nature of individuals with CP.

A prospective power analysis using G*Power (Faul et al., 2007) indicated that to achieve a medium effect size ($f = 0.25$) and acceptable power (0.8 ; $\alpha = 0.05$, one-tailed) for a repeated-measures design aimed at detecting between-group differences in cognitive bias performance (i.e. H1 and H2), the required sample size was 82. This power analysis was based on the primary between-subjects variable (groups, CP vs. NPC) and the primary within-subjects factor (stimulus type, pain/illness related vs non-pain/non-illness related). The present study was approved by the College of Health, Psychology and Social Care Research Ethics Committee, University of Derby (ethical approval no. ETH2021-4079).

2.2. Participants

Participants were recruited worldwide (response rate, 62.3%) via opportunity sampling. To recruit individuals with CP, online stratified sampling was employed.

Specific forums on social media sites such as Reddit (e.g. “r/Chronic Pain”), Facebook (“Pain Concern”, “Northamptonshire Chronic Pain Support”) and Health Unlocked (“Chronic Pain Forum”, “Osteoarthritis Action Alliance”) were used. Participants forming a Non Pain Control (NPC) group were recruited through a University Research Participation Scheme, and through social media including Reddit and Facebook. Eligibility criteria included: (i) ≥ 18 years old; (ii) fluency in English; (iii) normal or corrected-to-normal vision; (iv) experiencing pain for ≥ 12 weeks (CP group); and (v) not experiencing pain or not suffering from a pain/illness-related condition (NPC group).

Initially, a total of 153 participants were recruited. Of these, 77 participants were assigned to the CP group (mean age $\pm$ SD, 39.09 ± 16.75 years; female-to-male ratio, 9:1). All CP group participants had experienced pain for >12 weeks, with 86% reporting having also received a clinical diagnosis. The three most common CP conditions reported were fibromyalgia (19.5%), chronic headache/migraine (13.0%) and osteoarthritis (10.4%). A total of 34% of all individuals in the CP group reported multimorbidity (e.g. fibromyalgia with chronic headache). A total of 76 participants were assigned to the NPC group (mean age $\pm$ SD, 28.07 ± 9.69 years; female-to-male ratio, 3:2).

Within the CP group, the largest proportion of participants identified as American (54.2%), followed by British (38.9%) and Canadian (6.9%). In contrast, the NPC group was most commonly British (52.1%), followed by American (39.6%) and Canadian (8.3%). To ensure adequate cell sizes for chi-square analysis, nationality categories with fewer than four participants were excluded. Analysis therefore focused on the following nationalities: British, American and Canadian. A chi-square test of independence indicated that these differences in nationality distribution between CP and NPC groups were not statistically significant, $\chi^2 (2, n = 120) = 2.48, P = 0.29$.

With respect to the 1-month follow-up, 104 participants (68%) were retained, with 52 participants remaining in the CP group (mean age $\pm$ SD, 40.37 ± 16.47 years; female-to-male ratio, 10:1) and 52 remaining in the NPC group (mean age $\pm$ SD, 28.29 ± 9.96 years; female-to-male ratio, 2:1).

2.2.1. Demographic questionnaire

All participants were asked to state their age, sex, education level, first language, history of anxiety and/or depression, experience of pain for ≥ 12 weeks and diagnosis of a pain/illness-related condition or syndrome. Participants who answered “yes” to experiencing pain for ≥ 12 weeks were also asked to estimate the duration of their pain (in months and years).

2.2.2. Depression, Anxiety and Stress Scale-21 (DASS-21)

To ascertain whether IB and MB were due to pain rather than generalised anxiety/depression symptomatology, the DASS-21 (Henry & Crawford, 2005) was used. The DASS-21 is a 21-item questionnaire comprising three sub-scales: depression, anxiety and stress (seven items per sub-scale). Participants rate each item on a 4-point Likert-type scale ranging from 0 (“does not apply to me at all”) to 3 (“applies to me very much, or most of the time”), resulting in a maximum possible sub-scale score of 21 and a maximum total score of 63. Research has validated the psychometric properties of the DASS-21, finding each sub-scale to have adequate internal consistency, concurrent validity and Cronbach’s α values of 0.84, 0.74 and 0.79 for depression, anxiety and stress, respectively (Asghari et al., 2008; Musa et al., 2007; Wood et al., 2010). For the present study, the DASS-21 possessed excellent internal consistency ($\alpha = .94$).

2.2.3. Recent Pain Experiences Questionnaire (RPEQ)

To examine participants recent pain experiences four items derived from the Brief Pain Inventory Short-Form (Cleeland & Ryan, 1994) were used to create the RPEQ. Consistent with previous research (Heathcote et al., 2016; Said et al., 2019), participants were required to rate their: (i) Average pain intensity and worst pain intensity in the past 3 months from 0 (“No pain”) to 10 (“Worst pain possible”); (ii) the extent to which pain had interfered with daily activities over the past 3 months from 0 (“I don’t miss out on any activities”) to 10 (“I miss out on all activities”); and (iii) the frequency of their pain over the last 3 months from 1 (“ <1 day each month”) to 10 (“every day”). For the present study, the RPEQ possessed excellent internal consistency ($\alpha = .94$).

2.2.4. Experimental paradigms

2.2.4.1. IB—ambiguous scenarios task (AST). This task used the validated AST (Gaffiero et al., 2022). Participants were presented with 18 ambiguous pain scenarios designed to be equally interpretable as pain/illness related or non-pain/non-illness related. An example of such a scenario is “A ball hits you in the face. You look in the mirror and see your face is covered in ...”. This scenario is ambiguous as it allows for multiple interpretations; possible completions might include “mud” or “blood”. The task also included 18 additional ambiguous filler scenarios, intentionally designed to prevent pain priming. An example scenario is: “You get home from work and notice that you have left the ... on”, where

possible completions might include “light” or “oven”. Thus, the ambiguous “filler” scenarios had non-pain/non-illness-related associations (i.e. they were neutral in nature).

On each AST trial, participants were asked to read the scenario and imagine themselves in the situation. One practice trial was then completed by participants to ensure they understood the task requirements. Subsequently, the ambiguous pain and ambiguous neutral scenarios were presented randomly in the centre of the computer screen (font, Arial; size, 12) as shown below:

A ball hits you in the face. You look in the mirror and see your face is covered in ...

Participants were instructed to type the first word(s) that came to mind. This free-response format was chosen to enable participants to provide their own interpretations of the scenario.

The IB task was scored using the pre-defined coding scheme developed by Gaffiero et al. (2022). This objective, pre-defined approach ensured replicable scoring without the need for independent rater judgement. To expand, each participant's responses were matched against this coding list, with the word a participant provided for each scenario categorised as reflecting pain-related content (e.g. *hurt, ache, injury*) or non-pain-related content (e.g. *oven, light, parcel*). An IB index score is then calculated by subtracting the total number of pain-related completions from the number of non-pain-related completions. Scores can range from -18 to +18. A positive IB index score indicates a bias toward pain-related interpretations, whereas a negative score indicates a bias toward non-pain-related interpretations. A score of 0 reflects no bias.

2.2.4.2. MB—recall and recognition tasks

2.2.4.1.1. Surprise-free recall task. In this unexpected task, participants were given 3 minutes to type as many words as possible that they could remember having used to complete any of the scenarios (ambiguous pain or ambiguous neutral) from the AST into a blank box. The data were scored by counting the number of correctly recalled pain/illness-related and non-pain/non-illness-related solutions from the ambiguous pain scenario set. These counts were then converted into percentages to provide a measure of recall accuracy for both types of solutions (i.e. pain/illness related, and non-pain/non-illness related).

2.2.4.1.2. Delayed recognition bias task. This task was designed to present participants with 72 words (one at a time) comprising words they had previously generated

during the AST (“old” items; $n = 36$), and words they had not generated (“new” items, $n = 36$). Old items were derived from each participants' responses to ambiguous pain (18) and ambiguous neutral (18) scenarios in the AST. New items were selected using an individually tailored procedure to match, at the participant level, the number of pain/illness-related and non-pain related responses previously generated. For example, if a participant had produced 15 pain/illness-related responses and three neutral responses, they would be presented with 15 new pain/illness-related words and three new non-pain/illness related words. New words were drawn from the predefined pool of plausible word completions established in the validation study by Gaffiero et al. (2022), thereby ensuring lexical form and contextual relevance consistency across stimulus types.

Although individual words may vary in semantic properties such as frequency or concreteness, recognition performance assessed using Signal Detection Theory (SDT) accounts for differences in baseline familiarity by modelling hits and false alarms. Valence differences were intrinsic to the pain versus non-pain categorisations participants provided and therefore reflect the construct under investigation (rather than, for example, a methodological confound). All stimuli were presented in Arial font, size 12.

Participants were instructed that they would be presented with a series of words (one at a time) on the screen and asked to indicate whether they could remember using each word(s) to complete any of the ambiguous pain or neutral scenarios presented 1 month earlier in the AST. They were asked to respond “Yes” (“I did use this word(s)”) or “No” (“I did not use this word(s)”).

Recognition outcomes were calculated using SDT. “H” were defined as correctly identified old items, “M” as failing to identify old items, “FA” as incorrectly identifying new items as old, and “CR” as correctly identifying new items as new. SDT indices (d' and criterion C) were computed using proportional hit and false-alarm rates.

2.2.5. Procedure

The experiment was administered using Qualtrics (Provo UT). To limit priming, the study information was kept vague, and the study purpose was simply explained as an exploration of how ambiguous information can influence cognitive processing with two phases (now vs. 1 month later). Upon procurement of informed consent, participants were asked to complete the demographic questionnaire before completing the DASS-21. Next, participants completed the AST, followed by the RPEQ. To limit any issues associated with the recency

effect, the surprise-free recall task was the final task participants completed in the first phase. Upon completion of the surprise recall task, participants were partially debriefed—being simply reminded that they would be contacted 1 month later to take part in the second phase of the study which would involve making a series of judgements about a range of statements. This first phase of the study took ~30 minutes to complete.

Twenty-eight days later, participants were provided with a Qualtrics link to their personalised recognition task. For each personalised task, following informed consent, participants were provided with instructions for the delayed recognition task. Upon completion, participants completed the DASS-21 and RPEQ (provided after the recognition task to avoid priming) and were then fully debriefed and thanked for their time. On average, participants took 15 minutes to complete the second phase of the study.

2.2.6. Analytic strategy

To test hypotheses, the following analyses were conducted. First, group differences in demographic, psychological and pain-related variables were examined using independent-samples *t*-tests and χ^2 -tests. Second, an ANCOVA was conducted to assess whether individuals with CP interpreted more ambiguous scenarios as pain/illness-related compared to NPC's (H1), while controlling for relevant demographic and mood-related variables known to influence cognitive biases. Third, an ANCOVA was conducted to evaluate short-term recall of pain/illness-related solutions, with appropriate covariates included to account for individual differences (H2). Fourth, long-term recognition performance was assessed using signal detection theory (SDT) indices (d' and C), with independent-samples *t*-tests used to compare CP and NPC groups on pain- and non-pain related recognition outcomes (H3). Finally, partial correlation analyses were conducted separately within the CP and NPC groups to examine relationships between IB, recall bias and recognition performance (H4).

3. Results

3.1. Exploring between-group differences in the CP and NPC groups

An unpaired Student's *t*-test was carried out to investigate between-group differences in age, depression, anxiety, stress, average pain, worst pain, pain interference and pain frequency scores. A χ^2 -test was used to investigate sex differences (Table 1).

Significant between-group differences were found in age [$t(122.043)$, -4.992 ; $P \leq 0.001$; 95% CI, -15.39 ,

Table 1. Demographic characteristics for CP and NPC groups.

Variable	Group	
	CP ($n = 77$), mean $\pm$ SD	NPC ($n = 76$), mean $\pm$ SD
Age, years	39.10 $\pm$ 16.75	28.06 $\pm$ 9.68
Depression, score	9.56 $\pm$ 5.95	6.54 $\pm$ 5.73
Anxiety, score	6.88 $\pm$ 4.61	4.50 $\pm$ 3.62
Stress, score	9.21 $\pm$ 5.01	6.84 $\pm$ 4.57
Average pain, score	5.37 $\pm$ 2.04	1.46 $\pm$ 1.64
Worst pain, score	8.16 $\pm$ 1.64	3.14 $\pm$ 2.47
Pain interference, score	6.01 $\pm$ 2.45	1.21 $\pm$ 1.82
Pain frequency, score	5.42 $\pm$ 1.20	1.84 $\pm$ 1.12

CP: chronic pain; NPC: non-pain control; N: number of participants.

-6.65], depression [$t(143)$, -3.105 ; $P = 0.001$; 95% CI, -4.93 , -1.09], anxiety [$t(138)$, -3.357 ; $P \leq 0.001$; 95% CI, -3.78 , -0.97], stress [$t(145)$, -2.985 ; $P = 0.002$; 95% CI, -3.93 , -0.79], average pain [$t(145.083)$, -13.058 ; $P \leq 0.001$; 95% CI, -4.50 , -3.32], worst pain [$t(130.441)$, -14.755 ; $P \leq 0.001$; 95% CI, -5.69 , -4.35], pain interference [$t(140.389)$, -13.727 ; $P \leq 0.001$; 95% CI, -5.49 , -4.11] and pain frequency [$t(151)$, -19.038 ; $P \leq 0.001$; 95% CI, -3.96 , -3.21].

The χ^2 -test revealed a significant association between pain status and sex (χ^2 , 21.320; $P < 0.001$), indicating more females in the CP group ($n = 65$ vs. $n = 43$) and more males in the NPC group ($n = 31$ vs. $n = 6$). Taken together, these analyses indicate that compared with the NPC group, the CP group were of older age, mainly females and had higher depression, anxiety, stress, average pain, worst pain, pain interference and pain frequency scores.

3.2. Exploring IB between CP and NPC groups

To investigate H1, which posited that participants in the CP group would interpret more ambiguous scenarios in a pain/illness-related manner compared with NPCs, an ANCOVA was conducted. Prior to analysis, demographic (age, gender) and mood-related (stress, anxiety and depression) variables were screened. Z-scores revealed no outliers ($+/- 3$). Linearity was checked using Pearson's *r* correlations revealing age ($r = -0.265$, $P = <0.001$), depression ($r = 0.277$, $P = <0.001$), anxiety ($r = 0.264$, $P = <0.001$) and stress ($r = 0.216$, $P = 0.004$) all had a significant relationship with IB index (gender; $r = -0.046$, $P = 0.290$). As such, these variables were inserted as covariates. Levene's Test was non-significant indicating the homogeneity of variance assumption was met ($P = 0.15$).

The ANCOVA revealed no significant between-groups differences in IB index when demographic and mood-related variables were inserted as covariates, $F(6, 127) = 1.37$, $P = 0.245$, partial $\eta^2 = 0.011$.

3.3. Exploring memory recall bias between CP and NPC groups

To investigate H2, which posited that participants in the CP group would recall more pain/illness-related solutions generated during the IB task compared to the NPC group, an ANCOVA was conducted.

To account for variability in the number of solutions generated by participants, recall performance was expressed as category-specific recall rates rather than raw counts. Specifically, recall rates were calculated separately for (i) pain/illness-related and (ii) non-pain-related solutions. This process involved dividing the number of solutions correctly recalled by the total number of solutions generated for each specific category during the IB task. A memory recall bias index (MRB index) was then computed by subtracting the non-pain recall rate from the pain-illness recall rate. This provided a standardised measure of preferential recall (e.g. positive values represent a recall bias favouring pain/illness solutions, negative values represent a recall bias favouring non-pain/illness solutions, 0 represent no bias), while controlling for individual differences in solution generation.

Prior to analysis, demographic (age, gender) and mood-related (stress, anxiety and depression) variables were screened. *Z*-scores revealed no outliers ($+/- 3$). Linearity was checked using Pearson's *r* correlations revealing only age to be significantly associated with MRB index ($r = -0.353$, $P = <0.001$). As such, age was inserted as a covariate. Levene's Test was non-significant indicating the homogeneity of variance assumption was met ($p = 0.20$).

The ANCOVA revealed no significant between-groups differences in MRB index when age was inserted as a covariate, $F(2, 153) = 1.469$, $P = 0.227$, partial $\eta^2 = 0.227$.

3.4. Exploring memory recognition bias between CP and NPC groups

To test hypothesis 3, that participants in the CP group would correctly recognise more pain/illness-related solutions they generated in the ambiguous scenarios task compared with the NPC group, SDT metrics were calculated. Specifically, *d*-prime (*d'*) and Criterion C (*C*) were calculated to assess recognition performance. Table 2 provides descriptive statistics for Hits, Misses, False Alarms and Correct Rejections by group (CP, NPC) and stimulus type (pain, non-pain).

d' measures the ability to distinguish between pain/illness-related and non-pain/non-illness-related scenarios. This was calculated as the difference between the *Z*-scores of the *H* rate (proportion of correctly

Table 2. Descriptive statistics for signal detection outcomes (hits, misses, false alarms and correct rejections) for pain and non-pain stimuli by group (CP and NPC).

Variable	Group	
	CP ($n = 52$), mean $\pm$ SD	NPC ($n = 52$), mean $\pm$ SD
Pain hits	44.21 $\pm$ 27.80	51.70 $\pm$ 24.17
Pain misses	55.79 $\pm$ 27.81	48.30 $\pm$ 24.17
Pain false alarms	24.86 $\pm$ 26.24	21.61 $\pm$ 15.47
Pain correct rejections	75.14 $\pm$ 26.25	78.55 $\pm$ 15.39
Non-pain hits	41.98 $\pm$ 26.42	38.05 $\pm$ 25.25
Non-pain misses	57.54 $\pm$ 26.33	61.95 $\pm$ 25.25
Non-pain false alarms	20.78 $\pm$ 24.17	13.80 $\pm$ 13.50
Non-pain correct rejections	78.74 $\pm$ 24.46	86.20 $\pm$ 13.50

identified scenarios) and the FA rate (proportion of incorrectly identified scenarios) using the following equation: $d' = Z(H) - Z(FA)$. An unpaired *t*-test was conducted to compare *d'* scores for scenarios interpreted in a pain/illness-related manner. Levene's test was significant ($p = .046$) as such equal variances were not assumed.

The *T*-Test revealed the NPC group (mean $\pm$ SD; 0.22 ± 0.85) showed significantly higher *d'* scores compared with the CP group (mean $\pm$ SD; -0.22 ± 1.25); $t(102)$, 2.081 ; $P = 0.040$, $d = 0.408$, *two-tailed*. The 95% CI for the difference in *d'* scores between groups was 0.020 – 0.854 . This indicates that the NPC group was more sensitive in recognising pain/illness-related solutions they had previously generated than the CP group. Specifically, the NPC group demonstrated higher accuracy in identifying their previous pain/illness-related solutions. This result suggests participants in the NCP group were better at distinguishing between pain solutions they had previously generated than those they had not.

With respect to *d'* for scenarios interpreted in a non-pain/non-illness-related manner, Levene's Test was non-significant ($P = 0.87$), thus equal variances were assumed.

An unpaired *t*-test revealed no significant differences between group, $t(102)$, 0.973 ; $P = 0.333$, $d = 0.191$, *two-tailed*. This result indicates that the above observed differences in *d'* scores were specific to the recognition of previously generated pain/illness-related information rather than a general difference in recognition ability.

C assesses the decision threshold of participants. It indicates whether a participant is biased towards a "yes" or "no" response tendency when uncertain. It is calculated using the following equation: $C = -0.5 \times [Z(H) + Z(FA)]$. An unpaired *t*-test was conducted to compare *C* scores for scenarios interpreted in a pain/illness-related manner. Levene's test was significant ($P = 0.002$) as such equal variances were not assumed. The *T*-test revealed no differences in *C* between the NPC and CP groups; $t(102)$, -1.648 ; $P = 0.103$, $d = -0.323$, *two-tailed*.

For the non-pain/illness-related C data, Levene's test was non-significant ($P=0.061$) hence equal variances were assumed. A *T*-Test again revealed no differences in C between the NPC and CP groups, $t(102)$, -1.587 ; $P=0.116$, $d=0.311$, *two-tailed*.

Taken together, these findings indicate that neither group membership (CP or NPC) was associated with a systematic shift in decision threshold (i.e. response tendency). These findings suggested that the observed d' scores group differences for scenarios interpreted in a pain/illness-related manner are not attributable to a general response tendency bias when uncertain.

3.5. Exploring correlations between pain outcomes, interpretation, recall and recognition

To test hypothesis 4, which posited significant associations among the CB, a series of partial correlations were performed. These correlations examine the relationships between the number of scenarios interpreted as pain/illness related in the IB task (IB pain %), the number of pain/illness-related solutions correctly recalled in the free recall task (free recall pain %), and signal detection categories focusing on "old" and "new" pain solutions presented in the recognition task [1 month later (H, M, FA, CR)]. The analysis was conducted on data from 104 participants who fully completed these measures.

Pearson r correlations (*two-tailed*) on the CP group data revealed no significant relationships between IB Pain %, MB Pain % and any of the SDT measures (all $P > 0.05$). Similarly, Pearson r correlations (*two-tailed*) on the NPC group data revealed no significant relationships between IB Pain %, MB Pain % and any of the SDT measures (all $P > 0.05$). Together, these findings indicate that cognitive biases across interpretation, recall and recognition may operate relatively independently, rather than reflecting a single, unified bias profile within CP.

4. Discussion

The present study examined CBs, including IB and MB, in individuals with CP compared with NPCs. Contrary to Hypothesis 1, no between group differences were found in interpretation of ambiguous scenarios once demographic and mood-related variables were controlled for. Hypothesis 2 was also not supported, as individuals with CP did not recall a greater proportion of pain-related solutions than NPCs. However, partial support was found for Hypothesis 3: individuals with CP demonstrated significantly lower sensitivity (d') in recognising pain-related solutions after a one-month

delay, despite no group differences in response tendencies when uncertain (C). Finally, Hypothesis 4 was not supported, as no significant associations were observed between IB index, recall bias and recognition performance in either group. Collectively, these findings suggest that CBs in CP do not manifest uniformly and may operate in a more dissociable, time-dependent manner than previously assumed.

4.1. Interpretation bias in chronic pain

Previous research has frequently reported negative IB in individuals with CP, with meta-analytic evidence indicating a moderate tendency for pain-related interpretations of ambiguous information relative to healthy controls (Schoth & Liossi, 2016). However, IB does not appear to manifest consistently across all pain populations or task contexts. For example, sentence-generation paradigms have demonstrated medium-to-large effects in individuals with chronic or frequent headache (O'Hara et al., 2022), whereas studies in chronic fatigue syndrome suggest that IB is more strongly linked to fear-avoidance beliefs and symptom-specific threat appraisal (Hughes et al., 2016).

In the present study, the absence of group differences in IB after covariate adjustment suggests that previously reported IB effects may be partly attributable to demographic and mood-related factors, rather than pain status per se. Although the CP sample was heterogeneous and the ambiguous scenarios were not condition-specific, the use of a free-response format ensured that interpretations were participant-generated rather than constrained by forced-choice options. That said, while this approach enhances ecological validity it may also increase variability and reduce sensitivity to subtle group effects. Nonetheless, these findings indicate that IB in CP may be context-dependent, emerging under conditions of heightened threat relevance or affective salience rather than reflecting a stable, pain-specific cognitive bias. This has clinical relevance given that CP interventions are rarely tailored to diagnostic subgroups (National Institute for Health and Care Excellence [NICE], 2021 NG193).

4.2. Recall bias and short-term memory

With respect to hypothesis 2, no evidence was found for preferential recall of pain/illness-related information in the CP group. This aligns with the mixed and inconsistent literature on recall bias in CP populations (O'Hara et al., 2022). While early studies suggested enhanced recall of pain-related material in CP (Pincus & Newman, 2001; Pincus et al., 1995), more recent work has failed

to replicate these effects across conditions such as chronic low back pain and chronic headache (Karimi et al., 2016; Schoth et al., 2018). Notably, findings from Busch et al. (2006) suggest that recall impairments in CP may be modality-specific, with poorer performance observed for pain-related pictorial stimuli but not verbal material. One interpretation is that individuals with CP may engage in cognitive avoidance of pain-related information, which could attenuate recall performance and obscure any underlying bias. In the present study, recall stimuli were derived from participants' own interpretations during the IB task. If these self-generated pain-related solutions were not closely aligned with participants' lived experiences, recall performance may have been further constrained. Overall, the present findings reinforce the view that recall bias in CP is neither robust nor universal and may depend on stimulus characteristics, task demands and individual coping strategies.

4.3. Recognition bias and long-term memory

In contrast to IB and recall, recognition performance revealed a specific and theoretically informative group difference. Individuals with CP demonstrated reduced sensitivity in recognising previously generated pain-related solutions after a one-month delay, despite comparable performance for non-pain solutions and no evidence of response bias (C). This pattern suggests that CP is associated with impaired discriminability of pain-related information over time rather than a general memory deficit or tendency to endorse pain-related responses when uncertain.

These findings are consistent with the Attentional Cost Hypothesis (ACH) (Grisart et al., 2007). In the ACH it is proposed that ongoing pain consumes attentional resources, thereby limiting cognitive capacity for encoding, consolidation and retrieval processes. Although encoding was not directly assessed, reduced recognition sensitivity following a prolonged delay is compatible with disrupted encoding and/or consolidation under conditions of sustained cognitive load. Importantly, if reduced recognition accuracy were attributable solely to time-related forgetting, comparable impairments would be expected in the NPC group; this was not observed.

In contrast, the Hypervigilance Account (HA) (Mazza et al., 2018) predicts enhanced processing and memory for pain-related stimuli due to increased attentional allocation. The present findings do not support this account, as individuals with CP demonstrated lower, rather than higher, sensitivity to pain-related

stimuli and showed no evidence of a liberal response criterion.

Taken together, the recognition bias results align more closely with the ACH, suggesting that pain-related cognitive load may impair long-term memory processes rather than enhance sensitivity to threat.

4.4. Interactions between cognitive biases

Contrary to Hypothesis 4, no significant associations were observed between IB, recall bias and recognition outcomes in either group. This contrasts with findings from chronic headache populations, where co-occurring IB and recall biases have been reported (Schoth et al., 2018). The results are however consistent with evidence from non-clinical samples showing no relationships between IB and MB (Xu et al., 2024). The absence of associations in the CP group may reflect the complex cognitive demands imposed by persistent pain, which could disrupt or decouple the relationships between early-stage interpretation processes and later memory outcomes. In sum, the findings of the present study suggest that CBs in CP may not reflect a single, unified bias system, but instead operate as partially independent processes that vary across cognitive stages and over time. This highlights the importance of examining CBs longitudinally and within a temporally sensitive framework.

4.5. Theoretical implications

Existing models, including the Threat Interpretation Model (TIM; Todd et al., 2015) and the Integrated Functional Contextual Framework (IFCF; Van Ryckeghem et al., 2019), provide valuable insights into cognitive-affective mechanisms in CP, but offer limited predictions regarding MBs or their temporal dynamics. Considering the present findings, we propose the Integrated Pain Bias Framework (IPBF; see Figure 2) as a heuristic extension that explicitly incorporates MBs and distinguishes between early interpretative processes and later memory outcomes. The IPBF is intended as a provisional, testable framework to guide future longitudinal research rather than as a definitive explanatory model.

5. Limitations and future research

The present study has several limitations which should be acknowledged. First, recall was assessed shortly after the IB task, following the completion of questionnaires to minimise recency effects, whereas recognition was assessed after a one-month interval. As such, differences across tasks may reflect not only distinct memory

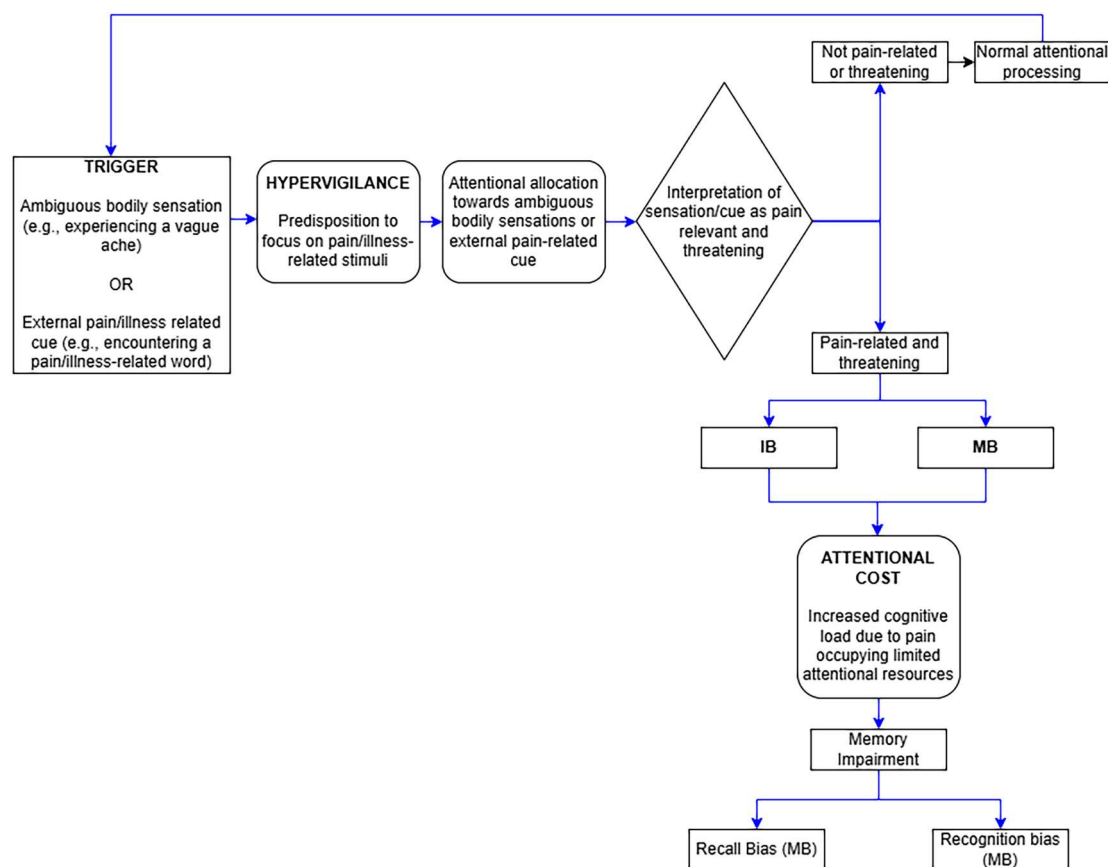


Figure 2. The integrated pain bias framework (IPBF). IB: interpretation bias; MB: memory bias.

processes (recall vs. recognition) but also the effects of time-related forgetting. Reduced recognition accuracy among individuals with CP may partly reflect memory decay rather than a specific impairment associated with cognitive bias. However, if time-related forgetting alone accounted for this pattern, comparable reductions would be expected across both groups; this was not observed. A second limitation concerns potential confounding variables arising from between-group differences in participant characteristics. The CP group was older and predominantly female relative to the NPC group, although these variables were statistically controlled where appropriate. It is also plausible that the groups differed on factors beyond pain status, such as sleep disturbance, which is known to impair cognitive functioning, including memory retrieval (Bothelius et al., 2019; Mazza et al., 2018). Moreover, the heterogeneous nature of the CP sample should also be considered when interpreting the findings. Due to insufficient statistical power, condition-specific subgroup analyses were not feasible and were therefore not conducted.

Third, the present study relied on self-report and behavioural paradigms without incorporating objective neuropsychological measures or socio-economic indicators. Although the tasks employed offer ecologically

relevant measures of IB and MB, the absence of formal cognitive testing limits conclusions regarding baseline memory functioning in CP populations. Finally, while retention at the one-month follow-up was high and balanced across groups (CP $N=57$; NPC $N=54$), it is possible that recognition performance in the CP group partly reflects variability in engagement (e.g. those with greater CP may have been more likely to discontinue onto phase 2).

Considering these limitations, future research should aim to more closely match participants on key demographic and cognitive variables, including age, gender, socio-economic status, baseline cognitive ability and/or baseline sleep quality, as well as incentivise all participants to complete all study phases, to limit attrition. Such approaches would allow the effects of CP on cognitive biases and memory processes to be more precisely isolated. In addition, the temporal dynamics of CBs warrant further investigation. Although the present study included a one-month follow-up, it did not examine how biases may develop, interact, or change over extended time-periods. Extended longitudinal designs examining how early attentional or interpretative processes influence later memory outcomes may offer valuable insight into the cognitive mechanisms

underpinning CP. This could then inform the development of more targeted intervention strategies.

In this context, emerging intervention approaches aimed at modulating cognitive and attentional processes may be particularly relevant. For example, recent evidence suggests that neurofeedback interventions show promise in chronic pain management by targeting neural mechanisms implicated in cognitive and affective regulation, which may indirectly influence cognitive biases associated with pain processing (Diotaiuti et al., 2024). Although preliminary, such approaches highlight the potential translational value of experimentally characterising cognitive biases in CP populations to inform the development of more targeted, mechanism-driven interventions.

6. Conclusion

In conclusion, this study examined CBs in individuals with CP compared with NPCs. While no group differences were observed in IB or short-term recall, individuals with CP showed reduced long-term recognition sensitivity for pain-related information. No associations were found between IB and MB in either group, indicating that these processes may operate independently rather than as a unified bias system. Overall, the findings suggest that CP selectively disrupts later-stage memory processes, consistent with the ACH. The proposed IBPF (Figure 2) offers a testable approach for understanding the temporal associations and dissociation between IB, recall and recognition processes in CP.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Generative artificial intelligence

In accordance with Taylor & Francis AI policy, we confirm that ChatGPT (GPT-5, OpenAI, December 2025 version) was used during the revision of this manuscript solely to support language clarity and editing. The tool did not contribute to the study design, data analysis, interpretation of results, or substantive content. All text was critically reviewed and edited by the lead author (DG), who takes full responsibility for the manuscript.

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Data availability statement

The dataset generated during the current study are available in the Open Science Framework (OSF) repository: https://osf.io/x2zk8/overview?view_only=6d1c68b9ff6c40a18d02e4786b65cc8a.

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